Ziyi Huang

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SUMMARY

Seeking a full-time software engineer position.
Results-driven software engineer with 4+ years of experience across internships, research, and projects, including **web applications with Java**, **recommendation systems**, and **entrepreneurial practice in software development**. Proficient in Java, Python, SQL, HTML, CSS, and JavaScript, with strong programming, problem-solving, and analytical skills.

EDUCATION

New York University Sep. 2023 – May 2025
M.S. in Computer Engineering; NYU Engineering Scholarship, GPA: 3.9/4.0 *New York, NY*

CUNY - City College Aug. 2019 – Dec. 2022
B.S. in Computer Science; Dean's List, Research Fellowship, GPA: 3.9/4.0 *New York, NY*

PROFESSIONAL EXPERIENCE

Software Engineer May 2025 – Present
Credit Acceptance *New York, NY*

- Integrated RouteOne and Dealertrack dealer management systems (DMS) into the Credit Approval Processing System (CAPS) as part of the Aggregators team to support loan origination workflows handling **100K**+ contracts per quarter
- Implemented industry-standard automated contract validation service to verify financial details (taxes, fees, and ancillary products) on RouteOne contract worksheets match approved deal structure, ensuring regulatory compliance while **reducing friction for franchise dealers to increase market share**
- Re-architected the contract validation workflow into a three-stage pipeline that transforms DMS requests into internal objects, applies DMS-agnostic validation logic against approved deal structures, and transforms back to DMS responses, **reducing DMS integration time and costs** by isolating DMS-specific protocols into reusable transformation layers
- Built a Business Process Tracker microservice to consume inbound requests via **AWS SQS** and persist them in segregated **AWS RDS** databases (separate PII/non-PII access controls) for request replay during downstream failures
- Collaborated with QA and Cloud Engineering teams to deploy **Spring Boot microservices** to production on **AWS EKS** via **GitHub CI/CD** pipelines, successfully delivering a GA release to **10K**+ quarterly active dealers nationwide

Software Engineer Intern Jun. 2022 – Aug. 2022
Cantor Fitzgerald *New York, NY*

- Accelerated crypto trading capabilities for Helix Financial by developing a real-time market monitoring system to expand the multi-billion-dollar institutional trading platform into digital assets
- Engineered an interactive crypto dashboard (**React.js**) for **300**+ institutional traders to monitor exchange metrics (trust scores, 24h trading volumes) across **50**+ supported crypto exchanges
- Architected RESTful API endpoints (**Java**, **Spring Boot**/**MVC**) to fetch and process crypto exchange data from CoinGecko API for dashboard consumption
- Optimized verification algorithm to ensure low-latency filtering of supported crypto exchanges for trading transactions

Software Engineer Researcher Jun. 2021 – Apr. 2022
City University of New York *New York, NY*

- Led the development of a 3D missile simulation program (**Unity3D**, **C#**) to enhance DoD's missile target tracking accuracy by generating 3D trajectory ground truth data
- Executed complete **Software Development Life Cycle** (SDLC) while adapting Agile methodologies from Scrum to Kanban to optimize team productivity amid changing academic schedules
- Parsed and transformed (**Python**) 1000+ rows of data from JSON to CSV format for trajectory validation
- Collaborated with validation team to verify data accuracy through camera-based calculations to achieve 98% precision in ground truth annotations

PROJECTS

NexEvent: Java Web Service Development - Personalized Event Recommendation Jan. 2024 – Dec. 2024

- Designed an interactive web page (**React.js**, **HTML**, **CSS**, **JavaScript**) for personalized event recommendations
- Implemented a web service (**Java**, **Spring Boot**) and 9+ **RESTful APIs** to handle HTTP requests and responses
- Designed a content-based recommendation algorithm to improve event recommendation based on favorite records
- Built relational database (**MySQL**) to store user data for consistency and migrated event data to NoSQL (**MongoDB**) for better scaling
- Deployed the web service seamlessly using **Docker** to **AWS EC2** to handle 150 QPS tested by **Apache JMeter**